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Surface-Confined Protection Stabilizes Pre-Annealing Crystallization for Ambient Blade-Coated Perovskites

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Abstract

Scalable manufacturing of perovskite solar cells is fundamentally limited by the vulnerability of perovskite crystallization to ambient moisture and oxygen, particularly during blade coating where an extended pre-annealing interval exposes unstable intermediates. Here, we introduce a surface-confined protection strategy to intrinsically stabilize perovskite film formation under ambient conditions. By introducing dipropylammonium trifluoroacetate (DPTA) into the perovskite precursor ink to spontaneously form a dense and self-assembled surface layer, selectively shielding the wet perovskite pre-film from environmental attack during the critical pre-annealing stage. This transient yet effective barrier preserves the $\text{PbI}_2 \cdot \text{NMP}$ intermediate to prevent pre-annealing degradation of the perovskite lattice even at high humidity. Simultaneously, the multifunctional ionic nature of DPTA allows strong coordination and hydrogen-bonding interactions with the perovskite lattice, leading to reduced bulk and interfacial defects. As a result, air-processed blade-coated MA-free perovskite solar cells reach an efficiency of 26.14% (certified at 25.75%), and retain 93.11% of the initial efficiency after 1,300 h under continuous 1 sun illumination tested at maximum-power-point. The strategy readily translates to manufacturing-relevant perovskite solar modules, delivering 22.72%-efficiency on substrate area of $100 \times 100 \text{ mm}^2$. These results establish surface-confined protection as a general principle for scalable perovskite photovoltaics under ambient conditions.

Introduction

Metal halide perovskites have emerged as one of the most promising photovoltaic materials, combining exceptional optoelectronic properties with solution processability and rapidly improving power conversion efficiency (PCE)^{1,2}. Currently, laboratory-scale perovskite solar cells (PSCs) rival crystalline silicon in efficiency, positioning the technology as a strong candidate for next-generation photovoltaics³. In order to transition from laboratory-scale production to large-scale manufacturing, it is crucial to ensure that the PSCs can withstand ambient conditions and are compatible with scalable production techniques⁴. However, the pathway of PSCs to industrial manufacturing remains obstructed by a fundamental materials challenge of the pronounced sensitivity of perovskite formation to ambient moisture and oxygen⁵. This vulnerability is particularly acute for scalable coating techniques, where environmental exposure is unavoidable and tightly coupled to film crystallization dynamics⁶⁻⁹.

The technologies such as blade-coating are widely regarded as leading candidates for large-area manufacturing of perovskite devices¹⁰⁻¹², which serve as an important intermediate platform between laboratory fabrication and industrial scalable coating owing to their compatibility with continuous processing and high material utilization. Unlike spin-coating in laboratory-scale fabrication, blade-coating inherently involves a prolonged interval between solution deposition and thermal annealing¹³⁻¹⁵. During the pre-annealing stage, solvent-coordinated intermediates and partially organized inorganic frameworks dominate the perovskite film, rendering the system highly susceptible to environmental perturbation¹⁶⁻¹⁸. Moisture and oxygen ingress at this stage alter solvent coordination, destabilize key intermediates such as $\text{PbI}_2 \cdot \text{N-methyl-2-pyrrolidone}$ (NMP), and redirect crystallization toward unfavorable pathways^{18,19}. Crucially, these early-stage disturbances are not merely superficial, and these disturbances propagate through grain growth and phase evolution, generating irreversible structural disorder and electronic defects that cannot be fully remedied by subsequent annealing¹⁹. As a consequence, the efficiency and operational stability of PSCs are particularly compromised in large-area blade-coated method. Existing strategies to mitigate environmental sensitivity largely rely on external control of the processing

atmosphere, including inert-gas glovebox^{20,21} and industrial dry room^{22,23}. While these effective approaches substantially increase capital and operational costs, eroding the economic advantages of solution processing. Alternative process-centric strategies, such as accelerated solvent removal²⁴⁻²⁶, gas quenching^{18,27} and narrowly defined coating windows²⁸⁻³⁰, partially suppress ambient interference but impose strict operational constraints to limit throughput and scalability. From a manufacturing perspective, these solutions treat the symptoms rather than the root cause of the absence of intrinsic protection for the vulnerable wet perovskite pre-film during the pre-annealing stage.

A fundamentally different approach is to engineer the perovskite ink and crystallization pathway such that environmental tolerance is built into the material system itself^{31,32}. In particular, interfacial wetting^{33,34}, precursor-solution stability^{35,36}, solvent coordination^{37,38} and ambient-induced side reactions^{39,40} can jointly perturb intermediate-state evolution, thereby affecting nucleation, buried-interface quality and ultimately device performance. Prior studies have demonstrate that specific solvent-coordinated intermediates, particularly $\text{PbI}_2\cdot\text{NMP}$, play a decisive role in directing the formation of high-quality of $\alpha\text{-FAPbI}_3$ perovskite film⁴¹⁻⁴³. However, these intermediates are themselves highly sensitive to moisture, leading to rapid transformation into less favorable adducts under ambient conditions. An effective solution must go beyond moisture scavenging to provide immediate and selective protection of the perovskite pre-film surface at the moment of deposition, precisely when the crystallization pathway is most vulnerable.

In this study, we introduce a surface-confined protection concept to address this critical gap. By incorporating dipropylammonium trifluoroacetate (DPTA) into perovskite precursor ink, we enable the spontaneous formation of a dense and molecularly thin surface layer during blade-coating in air. This layer acts as a transient armor to selectively block moisture and oxygen at the film-air interface, stabilizing the underlying solvent-coordinated intermediates throughout the pre-annealing period. Simultaneously, the multifunctional ionic nature of DPTA allows strong coordination and hydrogen-bonding interactions with the perovskite lattice, leading to reduced bulk and interfacial defects. As a result, the perovskite crystallization proceeds along the desired

pathway even under high humidity, yielding perovskite film with reduced residual strain, suppressed defect and enhanced phase stability. The PCE of MA-free PSCs fabricated entirely under ambient conditions increases from 22.99% to 26.14% (certified at 25.75%) with excellent stability. Even without external encapsulation, the PSCs based on DPTA retain 93.11% of the initial efficiency after 1,300 h of continuous 1 sun illumination at maximum power point tracking (MPPT), while the control device keeps only 68.68% of the initial efficiency under the same conditions. In addition, the thermal stability of the unencapsulated DPTA-treated device sustains 91.75% of its initial efficiency heating at 85 °C for 1,200 h, but the control device maintains 71.59%. Furthermore, this strategy translates to large-area perovskite solar modules (PSMs) without additional process complexity, and achieves a high PCE of 22.72% on substrate area of $100 \times 100 \text{ mm}^2$.

Results

Formation mechanism and characterization of the perovskite films

The extended time window between deposition and annealing in blade-coating technique presents a major vulnerability. During this stage, ambient moisture and oxygen corrode the wet perovskite pre-film, altering the crystallization pathway and leading to crystal imperfection, which cannot remedy by thermal annealing post-treatment. The molecular structure of DPTA in the Supplementary Fig. 1a. Density functional theory (DFT) calculations provide direct atomic-scale insight into the interfacial interaction between DPTA and perovskite components. As shown in Supplementary Fig. 1b and c, an I...H-O hydrogen bond further stabilizes the adsorption, while a Pb-N coordination bond indicating strong chemical anchoring between the amine group of DPTA and undercoordinated Pb^{2+} sites. This multi-faceted interaction mechanism is key to enhancing the quality of perovskite crystal structure during crystallization pathway. To clearly trace the evolution of the intermediate phases, we employed a simplified precursor system comprising only FAI, PbI_2 , DPTA, N,N-dimethylformamide (DMF) and NMP. Supplementary Fig. 2 and Fig. 1a schematically compares the crystallization behavior of samples under nitrogen glovebox and high

humidity of 85% relative humidity (RH) conditions. Within the nitrogen glovebox, both the control and DPTA-treated (target) samples follow the expected crystallization pathway, initially forming the δ -FAPbI₃, which is crucial for its subsequent conversion to the photoactive α -FAPbI₃⁴⁴. In contrast, Fig. 1a reveals that the formation of δ -FAPbI₃ is completely suppressed for the control under high humidity in air, indicating that moisture and oxygen ingress thoroughly disrupt its initial crystallization. Strikingly, the characteristic signature of the δ -FAPbI₃ remains observable for the DPTA-treated sample even under high humidity. Thus, we propose that DPTA likely interacts with precursor components to establish an armor-like kinetic barrier at the perovskite pre-film interface. This barrier retards the permeation and reaction of moisture/oxygen, thereby securing a crucial time window for the formation of δ -FAPbI₃.

To gain deep understanding of the regulatory role of DPTA in the perovskite crystallization process, we systematically examined the crystallization pathways under different humidity conditions (nitrogen glovebox, 40% RH in air, and 85% RH in air). In-situ X-ray diffraction (XRD) analysis of the wet films reveals the key mechanism by enhancing the environmental stability using DPTA. Under nitrogen conditions, Fig. 1b and Supplementary Fig. 3a show that the control and DPTA-treated samples exhibit similar result, primarily forming the PbI₂·NMP intermediate phase and successfully transforms into the crucial δ -FAPbI₃. However, a significant divergence occurs in air at 40% RH (Fig. 1c and Supplementary Fig. 3b). For the control, the intermediate phase shifts predominantly to an FAPbI₃·2DMF adduct^{41,45}, and no δ -FAPbI₃ is detected. This shift indicates that ambient moisture and oxygen disrupted the crystallization pathway. In contrast, the DPTA-treated sample maintains a PbI₂·NMP dominated intermediate phase and continues to produce the δ -FAPbI₃ even at 40% RH. When humidity increases to 85% RH (Fig. 1d and Supplementary Fig. 3c), the control remains unable to form the δ -FAPbI₃, whereas the DPTA-treated sample still exhibits clearly observable δ -FAPbI₃ diffraction peaks despite some evolution in its intermediate phase. Additionally, DFT calculations were used to estimate the adsorption energies of H₂O and O₂ on perovskite precursor components (Supplementary Table 1). The results show that although the pristine DPTA molecule exhibits strong water adsorption (-0.824 eV) due

to its hygroscopic nature, the DPTA-treated perovskite precursor components (FAI + PbI₂) display notably weaker adsorption for both H₂O (-0.339 eV) and O₂ (-0.289 eV) compared to pristine perovskite precursor components (-0.704 eV and -0.526 eV). These results provide compelling evidence to demonstrate that the introduction of DPTA effectively stabilizes the early-stage crystallization against moisture/oxygen attack. This protective effect stems from its ability to stabilize specific precursor intermediate phases (PbI₂·NMP). Thus, DPTA secures a critical time window for the formation of δ -FAPbI₃ under unfavorable conditions, thereby effectively improving the crystallization quality of the perovskite film in high humidity environments. This hypothesis is further confirmed by XPS. In contrast to the control, the target sample exhibits a pronounced O 1s signal (Fig. 1e), which is attributed to DPTA. This result confirms the successful incorporation of DPTA into the perovskite film. As shown in Fig. 1f, depth-profiling XPS analysis on the target sample reveals that the intense O 1s signal is exclusively localized at the surface. These are consistent with the XRD results, thereby confirming that DPTA forms an armor-like layer, which effectively shields the perovskite pre-film from moisture and oxygen erosion. To investigate the interaction between DPTA and the constituents of perovskite precursor solution, Fourier-transform infrared spectroscopy (FTIR) was performed. As illustrated in Fig. 1g, a narrowing of C=O vibrational peak without a significant positional shift following the PbI₂ introduction. Corroborating recent report, the observed peak sharpening indicates a more concentrated vibrational energy distribution and a reduction in structural disorder, which implies the formation of a specific interaction⁴⁶. The underlying mechanism involves the C=O group, acting as a Lewis base to coordinate with Pb²⁺ ions. The stretching vibration peak of N-H in DPTA shifts from 3439.8 to 3381.7 cm⁻¹ mixing with FAI, as presented in Fig. 1h, indicating an interaction between FAI and the N-H group⁴⁷. Collectively, these findings confirm the multifaceted interaction between DPTA and perovskite precursor components. In addition, X-ray photoelectron spectroscopy (XPS) analysis further confirms the presence of a chemical interaction. As shown in Supplementary Fig. 4, the target sample exhibits a negative binding energy shift for both Pb 4f and I 3d peaks, which is consistent with the DFT calculations. XPS results and FTIR analysis

collectively demonstrate DPTA successful coordination with the perovskite lattice, which is conducive to reduce the defects in bulk/interface.

Morphologies and nonradiative recombination processes of fresh and aged perovskite films

The comparative scanning electron microscope (SEM) analysis of perovskite films before and after high humidity aging, which directly confirms that the armor-like surface layer formed by DPTA effectively inhibits physical degradation. The top-view and cross-sectional SEM images of fresh films are presented in Fig. 2a and c. Although the initial control and target films exhibit comparable grain sizes, the target film demonstrates enhanced structural ordering and better vertical alignment. XRD further supports the enhanced crystallinity of the DPTA-treated fresh perovskite film (Supplementary Fig. 5), which is responsible for the formation of effective intermediate phase. After aging for more than 200 hours in a high humidity environment ($> 60\%$ RH), the SEM images reveal severe the degradation of control film due to exposure to moisture and oxygen, as shown in Fig. 2b and Supplementary Fig. 6. Following erosion of the grain boundaries by moisture and oxygen, their edges become indistinct, pores gradually emerge and the grains progressive decompose. In contrast, the DPTA-treated perovskite film exhibits no significant degradation. Based on these observations, we conclude that moisture-driven degradation of perovskites involves the preferential corrosion of grain boundaries along with phase segregation, which leads to the formation of non-photoactive δ -FAPbI₃, ultimately resulting in grain isolation and decomposition (Fig. 2e). Interestingly, DPTA can form a protective armor-like layer on the surface (Fig. 2f), effectively shielding the grain boundaries from moisture and oxygen exposure, thereby enhancing the long-term stability.

Supplementary Fig. 7a shows that the control exhibits a pronounced increase in the O *1s* signal, indicating that moisture infiltration induced perovskite decomposition as well as the formation of new oxidized phases. Conversely, the O *1s* signal of target remains stable after aging under the same conditions (Supplementary Fig. 7b), demonstrating the exceptional effectiveness of DPTA surface layer in blocking moisture ingress and preventing oxidation. As evidenced by the results in Supplementary Fig. 8, the Pb *4f* peaks of the control gradually exhibit a negative shift

after the high humidity aging process, while the Pb $4f$ peaks of target keeps the same locations at the same aging conditions. The negative shift in Pb $4f$ binding energy suggests a change in the chemical speciation of lead, indicating a transformation of the local coordination environment from $[\text{PbI}_6]^{4-}$ octahedra to an oxygen-coordinated electron-rich state. This is consistent with the changes observed in the O $1s$ peaks of control film. Additionally, the I $3d$ peaks in Supplementary Fig. 9 also exhibit a negative shift, which is attributed to the altered chemical environment of iodide species as they dissolve from the decomposing lattice. These observations are consistent with the SEM results, effectively demonstrating that DPTA would form an armor-like layer on the surface to prevent moisture and oxygen infiltration. Photoluminescence (PL) mapping which scan area is $50\text{ }\mu\text{m} \times 50\text{ }\mu\text{m}$ was performed on the fresh control and target films, as shown in Fig. 2g and i. DPTA-treated film exhibits higher intensity compared to control one, indicative of suppressed nonradiative recombination. Fig. 2h and j show the PL mapping results of control and DPTA-treated samples after the same humidity aging. A pronounced difference in PL intensity is observed for the control before and after aging, whereas the target demonstrates remarkable stability with negligible change. Furthermore, the atomic force microscopy (c-AFM) measurements (Supplementary Fig. 10) reveal distinct differences between the control and DPTA-treated perovskite films before and after the same humidity aging. The fresh control film exhibits much lower local current and displays a dramatic decrease to near-zero current after humidity aging, evidencing severe degradation. In contrast, the fresh DPTA-treated film shows a substantially higher initial current with only minimal degradation after aging. Collectively, these findings provide strong evidences for the robust protective capability of DPTA, which is crucial for improving device stability.

Properties of control and DPTA-treated perovskite films

As determined by dynamic water contact angle measurements in Supplementary Fig. 11, the wettability of perovskite film on the ITO/SAM substrate slightly enhances after DPTA treatment. Atomic force microscopy (AFM) images are presented in Fig. 3a and b, the surface roughness of DPTA-treated perovskite film reduces from 34.4 to 31.0 nm, which means that DPTA treatment

facilitates the formation of smooth and dense perovskite film. The effect of DPTA treatment was further checked by using the PL spectra, as shown in Fig. 3c. The emission intensity of DPTA-treated perovskite film is more than twice that of the control sample, indicative of the suppressed nonradiative recombination, which contributes to the high open-circuit voltage (V_{oc}) in the modified devices. Fig. 3d shows the time-resolved PL (TRPL) spectra of control and DPTA-treated target perovskite films, and the fitting parameters are summarized in Supplementary Table 2. Based on the average carrier lifetimes (detailed calculated process in Supplementary Note 1), the target film exhibits a notably prolonged carrier lifetime relative to the control film, indicating that DPTA effectively reduces defects to suppress nonradiative recombination. In addition, we used space-charge-limited current (SCLC) measurements to quantify the defects within the perovskite films by electron-only (ITO/SnO₂/perovskite/PCBM/Ag) and hole-only (ITO/NiO_x/perovskite/spiro-OMeTAD/Au) structures. Using the trap-filled limit voltage (V_{TFL}) extracted from the dark current-voltage (I - V) characteristics (Supplementary Fig. 12), the defect density was obtained (detailed calculation in Supplementary Note 2). The comparison results are summarized in Fig. 3e. The control perovskite film shows significantly high defect density, especially for electron defect ($2.20 \times 10^{15} \text{ cm}^{-3}$). In contrast, the DPTA-treated perovskite film with relative low values of $0.58 \times 10^{15} \text{ cm}^{-3}$ for electron defects and $1.05 \times 10^{15} \text{ cm}^{-3}$ for hole defects. These findings show that the DPTA treatment approach successfully reduces defect, which is in line with long carrier lifetime and improved crystallinity.

To deeply investigate the residual strain within the perovskite films before and after aging, the grazing incidence X-ray diffraction (GIXRD) measurement was performed to acquire the depth-dependent crystal structure information, as shown in Fig. 3f and Supplementary Fig. 13. The characteristic peak of the fresh control perovskite film gradually shifted to lower angles compared to fresh DPTA-treated film. The DPTA-treated perovskite film shows a relatively smooth slope, indicating that DPTA treatment reduces residual stress and obtain high-quality of perovskite films. Furthermore, for aged perovskite films, DPTA treatment exhibits a smaller slope change compared to fresh films (0.03), whereas the slope change of the control film is 0.04, further indicating DPTA

treatment effectively relax the residual strain, which is beneficial to enhanced stability.

Performance of perovskite photovoltaics

Based on the significant improvement in photoelectric property, crystallinity, strain and stability of the perovskite film after DPTA treatment, we fabricated PSCs with a structure of glass/ITO/SAM/perovskite/PDADI/PCBM/BCP/Ag (Supplementary Fig. 14). The influence of DPTA concentration (0, 1.0, 1.5 and 2.0 mg/ml) on device performance was systematically investigated. Photovoltaic parameters from 15 individual devices were statistically analyzed, including V_{OC} , fill factor (FF), short-circuit current density (J_{SC}) and PCE, as depicted in Supplementary Fig. 15 and Supplementary Table 3. DPTA-treated devices with optimal concentration of 1.5mg/mL not only achieve higher efficiency but also exhibit better reproducibility than the control ones. Fig. 4a presents current density-voltage (J - V) curves of the champion PSCs with and without DPTA treatment scanned at forward and reverse directions. Evidently, DPTA treatment endows the champion PCE of 26.14%, entailing a V_{OC} of 1.183 V, a J_{SC} of 26.14 mA cm⁻², and an FF of 84.52%, which outperforms that of the control device (22.99%). This performance was further validated by an independently certified PCE of 25.75% (Supplementary Fig. 16). The hysteresis index (HI, calculated according to the Supplementary Note 3) decreases markedly from 0.130 for the control device to 0.021 for the DPTA-treated device. This significant reduction between the forward and reverse scan directions originates from an effective improvement in crystallization quality, reduced density and relaxed residual strain by DPTA treatment. Fig. 4b exhibits the external quantum efficiency (EQE) spectra as a means of verifying J_{SC} . The integrated current values of control and target devices are 25.88 and 26.03 mA cm⁻², which is consistent with the current values obtained from J - V measurement. Supplementary Fig. 17 presents the steady-state power output of the DPTA-treated PSC at its maximum power point of 1.01 V, exhibiting a remarkable steady-state PCE of 25.89% for more than 150 s, identifying with J - V result.

The variation of V_{OC} and J_{SC} as a function of light intensity was examined, as shown in Fig. 4c and Supplementary Fig. 18. The V_{OC} fitted slope decreased from 1.67 to 1.18 kBT/q with DPTA

treatment, indicating suppressed nonradiative recombination. The relationship of the J_{sc} versus light intensity can be expressed by using the exponential factor α . The α value of DPTA-treated device is 0.991, higher than the control device (0.975), when α closer to 1, indicating that defect-assisted recombination is reduced⁴⁸.

We also fabricated perovskite films on a $100 \times 100 \text{ mm}^2$ substrate by using the DPTA-treated strategy. The large-area perovskite film was then subdivided into eight distinct regions (Supplementary Fig. 19) for small-area device fabrication to access the uniformity, ensuring each region was processed identically to form individual planar p-i-n devices. The performance across eight regions is highly consistent (Fig. 4d, Supplementary Fig. 20 and Supplementary Table 4), indicating the good uniformity of large-area DPTA-treated perovskite film. We fabricated PSMs with an active area of 54.51 cm^2 consisting of 15 single-cell connected in series, as shown in Supplementary Fig. 21. Fig. 4e shows the I - V curve of DPTA-treated PSM with photovoltaic parameters. The PCE histogram of DPTA-treated PSMs is provided in Supplementary Fig. 22, which was derived from six devices (Supplementary Table 5). The PCE of champion PSM reaches 22.72% with short-circuit current (I_{sc}) of 91.36 mA, V_{oc} of 17.38 V and FF of 78.00%, one of the highest efficiencies for this type of PSMs.

Stabilities of perovskite photovoltaics

To evaluate stability of PSCs with and without DPTA treatment, we explored the control and DPTA-treated target perovskite films aging at $> 60\%$ RH for 200 hours by ultraviolet-visible (UV-vis) spectroscopy (Supplementary Fig. 23). Initially, the light absorption exhibits stronger intensity compared with control film due to the more uniformity of DPTA-treated perovskite film. After aging at the same conditions, the control perovskite film experiences more significant degradation, whereas the target film maintains more stable light absorption. We also conducted XRD analysis to examine the evolution of the control device aging at $> 60\%$ RH, as shown in Fig. 5a and b. XRD analysis clearly reveals significant differences in phase stability between control and target perovskite films. The control decomposes rapidly, with the δ -FAPbI₃ decomposition phase appearing after just 1 day. By the fifth day, the intensity ratio of δ -FAPbI₃ to α -FAPbI₃ diffraction

peaks reached nearly 1:1, and by the ninth day, it exceeded 2:1. In contrast, the DPTA-treated perovskite film demonstrates exceptional stability, with δ -FAPbI₃ only becoming detectable on the ninth day. The corresponding photo of the control and DPTA-treated perovskite films are presented in Supplementary Fig. 24. For the control perovskite film, a yellow color appears at the edge after aging for 3 days, and obvious surface decomposition is observed after 20 days, with the edges undergoing decomposition. In contrast, the DPTA-treated perovskite film shows no significant changes aging for 20 days under the same conditions. After 60 days, the control film fully decomposes, while the target film shows decomposition only at the edges.

The maximum power point tracking (MPPT) results (Fig. 5c) demonstrate that the unencapsulated DPTA-treated device exhibits excellent stability, maintaining 93.11% of its initial efficiency after 1,300 hours of continuous 1 sun illumination. In contrast, the control device exhibits a 10% drop in efficiency within just 120 hours, and remains only 68.68% of its initial efficiency after 1,300 hours. Furthermore, we evaluated the thermal stability of the unencapsulated perovskite devices heating at 85 °C by using *J-V* testing (Fig. 5d). The DPTA-treated device sustains 91.75% of its initial efficiency for 1,200 hours, showing more stable efficiency than the control device (71.59%). Additionally, we stored devices under ambient with 30-50% RH to assess long-term environmental stability (Supplementary Fig. 25). After 1,000 hours, DPTA-treated device maintains 98.59% of its initial efficiency, while control devices remain 86.50% of its initial efficiency. These results further underscore the impact of DPTA on device stability owing to formation of a protective surface layer to effectively shield the device from degradation caused by moisture and oxygen.

Discussion

In summary, we demonstrate that the primary barrier to ambient blade coating for high-performance of MA-free PSCs originates from uncontrolled environmental interference during the pre-annealing stage of film formation, rather than from limitations of the annealed perovskite itself. By introducing a surface-confined protection strategy using DPTA, we decouple perovskite

crystallization from ambient moisture and oxygen through the spontaneous formation of a dense, self-assembled surface layer that transiently shields vulnerable solvent-coordinated intermediates. This protection preserves the desired crystallization pathway, suppresses defect formation and residual strain, and leads to perovskite films with both high electronic quality and exceptional phase stability. These synergistic effect enables the PCEs of PSCs and PSMs to increase 26.14% (certified at 25.75%) and 22.72% with remarkable long-term operational stability. Our results establish surface-confined protection during pre-annealing as a general materials design principle for scalable perovskite manufacturing.

Methods

Material

4PADCB was purchased from Li wei New Material Technology Co., LTD. Lead iodide (PbI_2 , >98.0%) and anddipropylammonium thrifluoroacetate (DPTA, >98.0%) were obtained from TCI. Cesium iodide (CsI , 99.999%) was purchased from Alfa Aesar (China). Formamidinium iodide (FAI, 99.9%), 1,3-Diaminopropane dihydroiodide (PDADI, 99.5%) and bathocuproine (BCP, 99%) were obtained from Xi'an Yuri Solar Co., Ltd. [6,6]-phenyl- C_{60} -butyric acid methyl ester (PC_{60}BM , 99.9%) were purchased from Advanced Election Technology (Canada). Dimethyl sulfoxide (DMSO >99.9%) and N,N - Dimethylformamide (DMF, >99.9%) were purchased from Aladdin. 1-Methyl-2-pyrrolidinone (NMP, $\geq 99.7\%$), 2-Methoxyethanol (2Me, $\geq 99.9\%$) and 2-Propanol (IPA, 99.9%) were purchased from Sigma-Aldrich. Chlorobenzene (CB) were purchased from Sinopharm (China).

Preparation of small area PSCs

25 × 25 mm ITO glasses (SuZhou ShangYang Solar Technology Co.,Ltd, TX11-12AR) were sequentially ultrasonicate cleaned by acetone and isopropanol. The cleaned substrates were dried with a air flow and Ar plasma for 2.5 min. The 4PADCB solution (1 mg/ml DMSO) was spin-coated onto the substrates at 3000 rpm for 30 s, followed by drying on a hot plate at 120 °C for 10

min. Perovskite precursor solution with a molar concentration of 1.2M was prepared elsewhere with a stoichiometric composition of $\text{FA}_{0.9}\text{Cs}_{0.1}\text{PbI}_3$ by adding FAI, CsI and PbI_2 in to mixed DMF/NMP solution (9.2:1). Different concentrations (0, 1.0, 1.5 and 2.0 mg ml^{-1}) of DPTA was added in the perovskite solution. After stirring for 2 h, 2ME (same as NMP) was added. Following this, for small-area devices 20 μL perovskite solution was blade-coated with a rate of 3mm s^{-1} in the air, the gap between blade and substrate is 220 μm and the pressure of the N_2 -knife is 0.4 MPa. Next, the films were transferred to a hotplate and annealed at 130 $^\circ\text{C}$ in air for 20 min. For the device with passivation, the substrates were transferred into N_2 glovebox to spin-coating PDADI (0.5 mg/ml IPA) with 3000 rpm for 30 s and annealed at 100 $^\circ\text{C}$ for 10 min. PCBM (20 mg/ml CB) was spin-coated as an electron transport layer (ETL) with 4000 rpm for 30 s. Next, the saturated solution of BCP in IPA was spin-coated on the PCBM layer at 5000 rpm for 30 s. Finally, Ag (150 nm) was thermally evaporated onto the samples using a vacuum deposition system.

Preparation of large-area PSMs

For large-area devices, the procedures largely mirror those used for small-area devices, with the primary adjustment being the rotational speed, which is modified to accommodate the large-area modules. The 4PADCB solution (1 mg/ml DMSO) was spin-coated onto the substrates at 2000 rpm for 30 s, followed by drying on a hot plate at 120 $^\circ\text{C}$ for 10min. As for large-area devices, the amount of perovskite solution was increased to 135 μL . PDADI (0.5 mg/ml IPA) was spin-coated with 2000 rpm for 30 s and annealed at 100 $^\circ\text{C}$ for 10 min. PCBM (18 mg/ml CB) was spin-coated as an ETL with 2000 rpm for 30 s. Next, the saturated solution of BCP in IPA was spin-coated on the PCBM layer at 2000 rpm for 30 s. Finally, Ag (100 nm) was thermally evaporated onto the samples using a vacuum deposition system. To fabricate modules, the ITO electrode, 4PADCB/perovskite/PDADI/PCBM/BCP stacking layer, and Ag electrode were ablated by laser scribe sequentially to form a P1-P2-P3 pattern. The laser process and calculation of the module's effective area are same with our previous report^{11,49}.

Device characterization

The J - V were measured in ambient air with an AM 1.5 G radiation generated by a solar simulator based on xenon lamps (Elitch, Class AAA Solar Simulator) and a Keysight B2901A Source-Meter. A standard crystalline silicon cell was used to calibrate the light intensity of solar simulator before measurement. The devices were measured with a scan rate of 0.01 V/s and a delay time of 1 ms with the reverse scan range from 1.2 V to -0.1 V and the forward scan from -0.1 V to 1.2 V. The dark J - V characteristics of the devices were also assessed using a Keysight B2901A Source-Meter with an only electron structure of ITO/SnO₂/perovskite/PCBM/Ag and an only hole structure of ITO/SAM/perovskite/spiro-OMeTAD/Au. The scanning rate employed was 0.003 V/s, covering a range from 0 V to 3 V. A Keithley 2460 Source-Meter was used to measure the I - V of large-area PSCs under simulated AM 1.5 G irradiation generated by a solar simulator based on xenon lamps (WACOM, Class AAA Solar Simulator). There was a 1 ms delay time and a 0.1 V/s scan rate. The light intensity of the solar simulator is calibrated using a silicon reference cell that has been certified by the National Institute of Metrology in China. FTIR spectrum was performed on FTIR (Nicolet IS10) with a wavenumber resolution of 0.5 cm⁻¹ in the range 400-4000cm⁻¹. UV-vis spectrophotometer (Hitachi UH-4150) was used to record UV-vis absorption spectra. With the assistance of an Enlitech QE (R3018) measuring device, the EQE spectra were acquired. The film morphologies were measured by using SEM (Hitachi SU8020). The AFM images was captured in contact mode using an atomic force microscope (Dimension Icon, BRUKER, USA). XPS measurements were carried out on an X-ray photoelectron spectrometer (Thermo ESCALAB 250Xi). PL was measured on a fluorescence spectrometer (Pico1000, Light-Stone CO, LTD) using the time-correlated single photon counting technique. TRPL and PL mapping, were recorded using a custom-built scanning confocal microscope equipped with a FLIMLABs 405 nm pulse laser, a galvo scanner, and an IDQ time-correlated single photon counter. The static water contact angles were measured by contact angle meter (DSA100). MPPT was carried out using PVL-6001M-32B.

DFT calculation

All the calculations were performed by Vienna ab initio simulation package (VASP) with the projector augmented wave pseudopotentials (PAW) to describe the interaction between atomic cores and valence electrons with density functional theory (DFT). The Perdew-Burke-Ernzerhof (PBE) functional within the generalized gradient approximation (GGA) was used to implement DFT calculations. The surfaces were modeled by a slab consisting of 2×1 periodicity in the a-b plane, and the reasonable vacuum layers were set around 15 Å in the z-directions to avoid interaction between planes. A cutoff energy of 400 eV was provided and a $5 \times 5 \times 1$ Monkhorst Pack k-point sampling was chosen for the well converged energy values. Geometry optimizations were pursued until the force on each atom fell below the convergence criterion of 0.01 eV/Å and energies were converged to within 10^{-5} eV. The van der Waals correction was employed for accurate structural relaxation.

Data Availability

All the main data are available in the main text and the Supplementary Information. All other data of this study are available from the corresponding authors on request. Source Data are provided with this paper.

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Author Contributions

L.D., X.Z. and D.Y. conceived the ideas. Z.Z., X.Z. and D.Y. supervised the project and progress.

L.D. and X.Z. designed the experiments. L.D. fabricated the devices and performed the main characterizations. R.Y. and S.Y. assisted in device fabrication and conducted PL mapping, AFM and GIXRD measurements. Z.L. offered suggestions on the DFT calculation strategy, and Z.Z. carried out the calculations. S.L provided guidance on the writing of the manuscript and L.D. wrote the first draft of the manuscript. All authors discussed the results and approved the manuscript.

Competing Interests

The authors declare no competing interests.

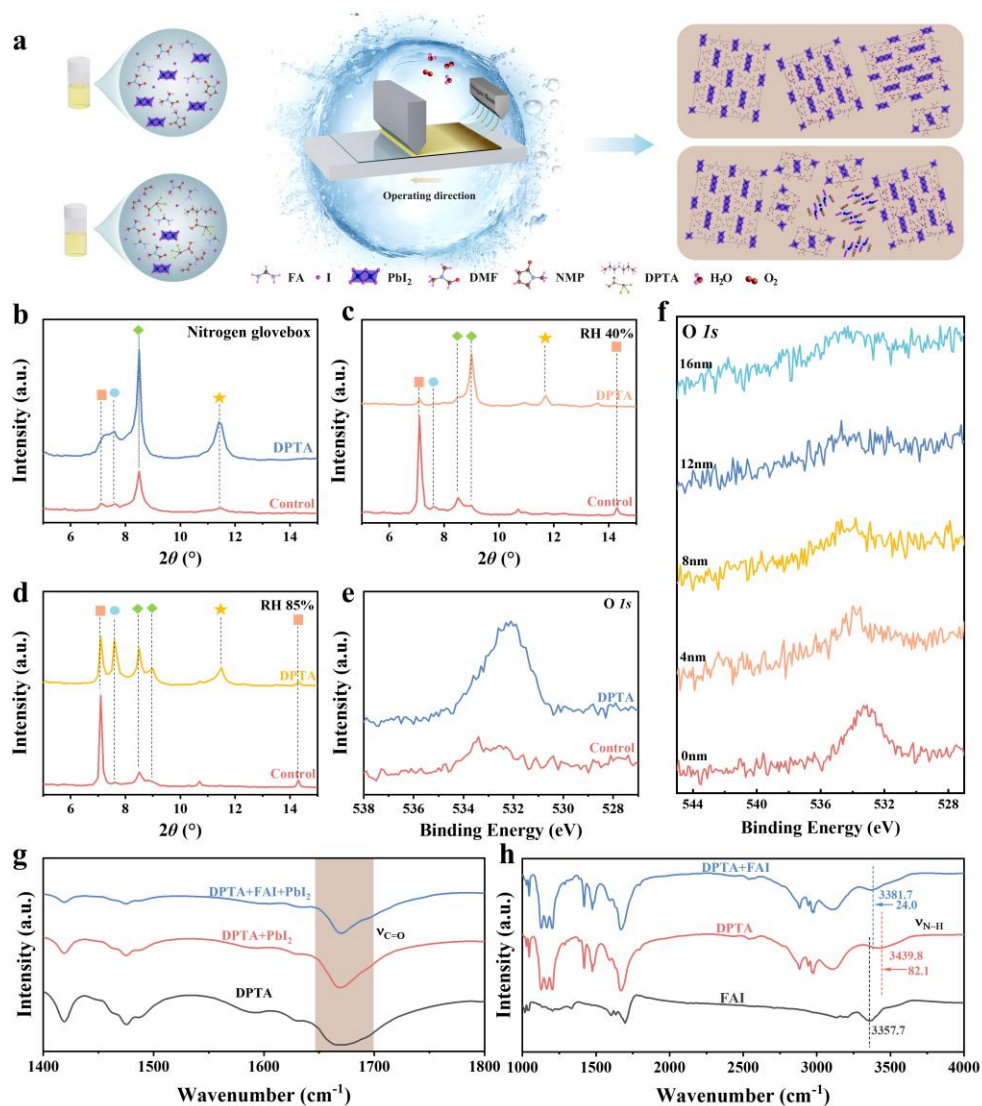


Fig. 1 | Formation mechanism and characterization of the perovskite films. **a** Schematic diagram for the formation of perovskite pre-film by blade-coating under high humidity in air. XRD patterns of wet films for control and DPTA-treated **b** in nitrogen glovebox, **c** at 40% RH and **d** at 85% RH. **e** XPS spectra of O 1s binding energy on the surface of control and DPTA-treated perovskite films. **f** Depth-profiling XPS of O 1s in DPTA-treated perovskite film. **g** FTIR spectra of DPTA, DPTA+PbI₂ and DPTA + FAI + PbI₂ films. **h** FTIR spectra of FAI, DPTA and DPTA + FAI films.

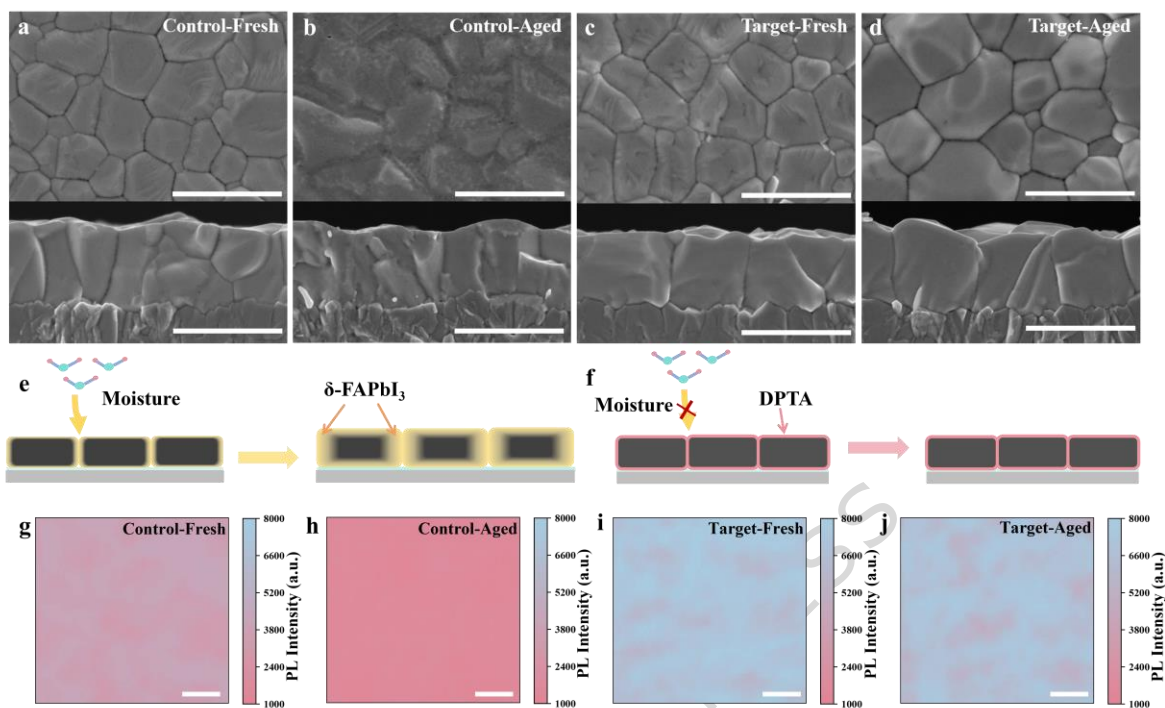


Fig. 2 | Morphologies and nonradiative recombination processes of fresh and aged perovskite films. Top-view and cross-sectional SEM images of the fresh **a** control and **c** DPTA-treated perovskite films, the aged **b** control and **d** DPTA-treated perovskite films, respectively. Scale bars represent 1 μm . Schematic illustrations depict the aging processes of **e** the control and **f** the DPTA-treated perovskite films. PL mappings of **g** the fresh control film, **h** the aged control film, **i** the fresh target film and **j** the aged target film. Scale bars represent 10 μm .

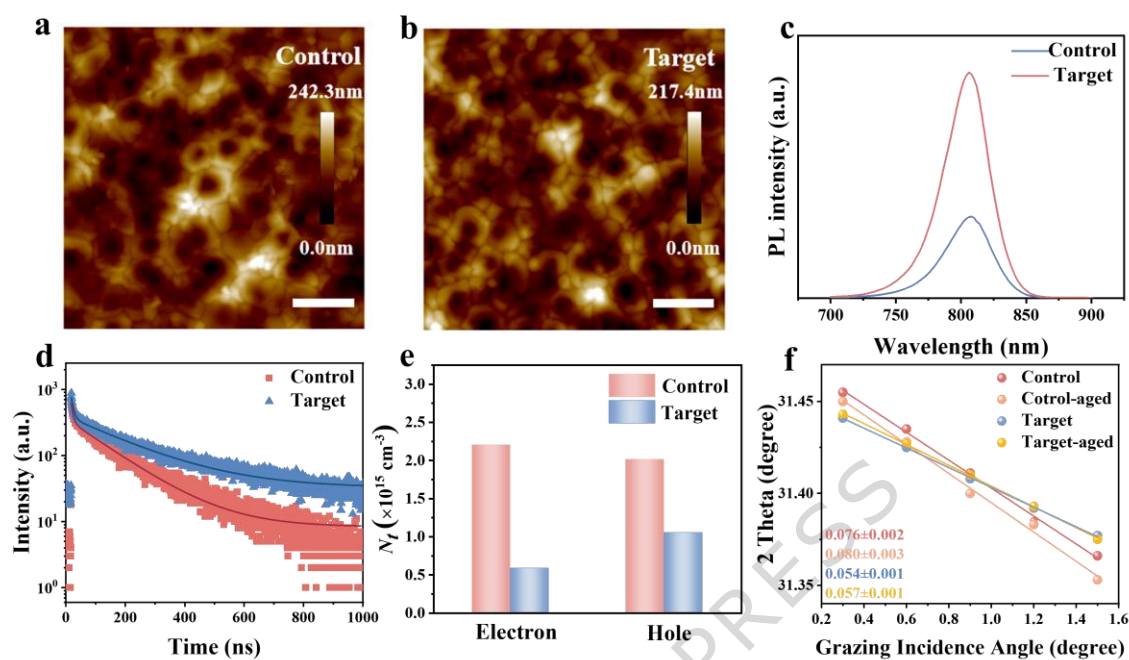


Fig. 3 | Properties of control and DPTA-treated perovskite films. AFM images of the **a** control and **b** DPTA-treated perovskite films. Scale bars represent 1 μm . **c** Steady-state PL spectra, **d** TRPL spectra and **e** defect densities of the control and DPTA-treated perovskite films. **f** Residual strain distribution of the control and DPTA-treated perovskite films before and after aging.

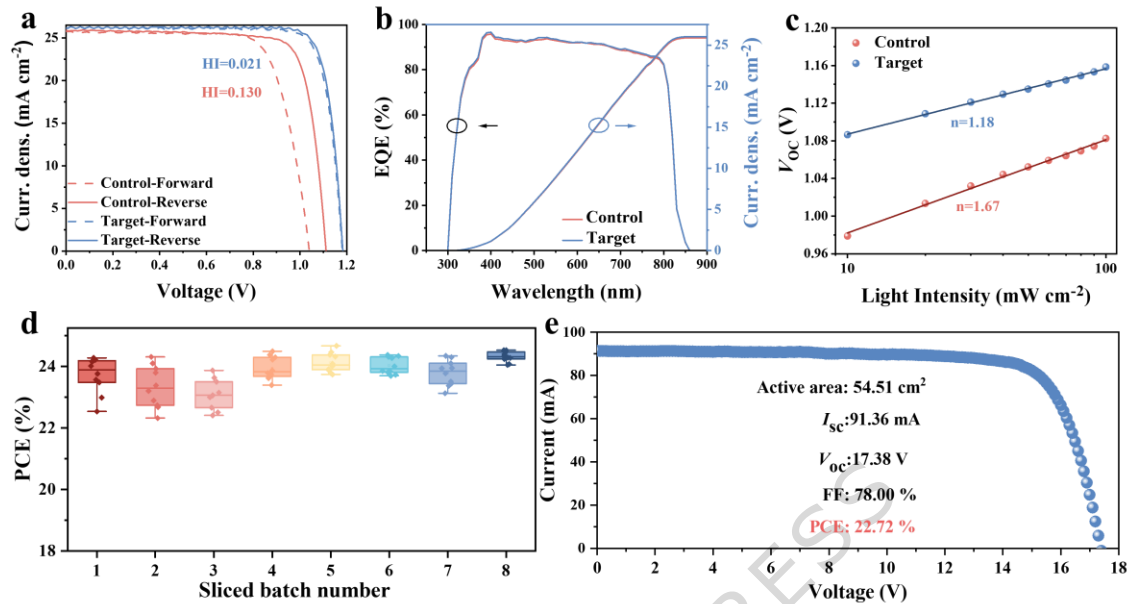


Fig. 4 | Performance of perovskite photovoltaics. **a** J - V curves of control and DPTA-treated PSCs measured by forward and reverse directions. **b** EQE and integrated photocurrent density of the control and DPTA-treated devices. **c** V_{OC} versus light intensity curves for PSCs with and without DPTA treatment. **d** Distribution of PCEs for cutting slices from large-area DPTA-treated perovskite film. Error bars represent the standard deviation calculated from 10 independent devices. **e** I - V curves of the champion PSMs.

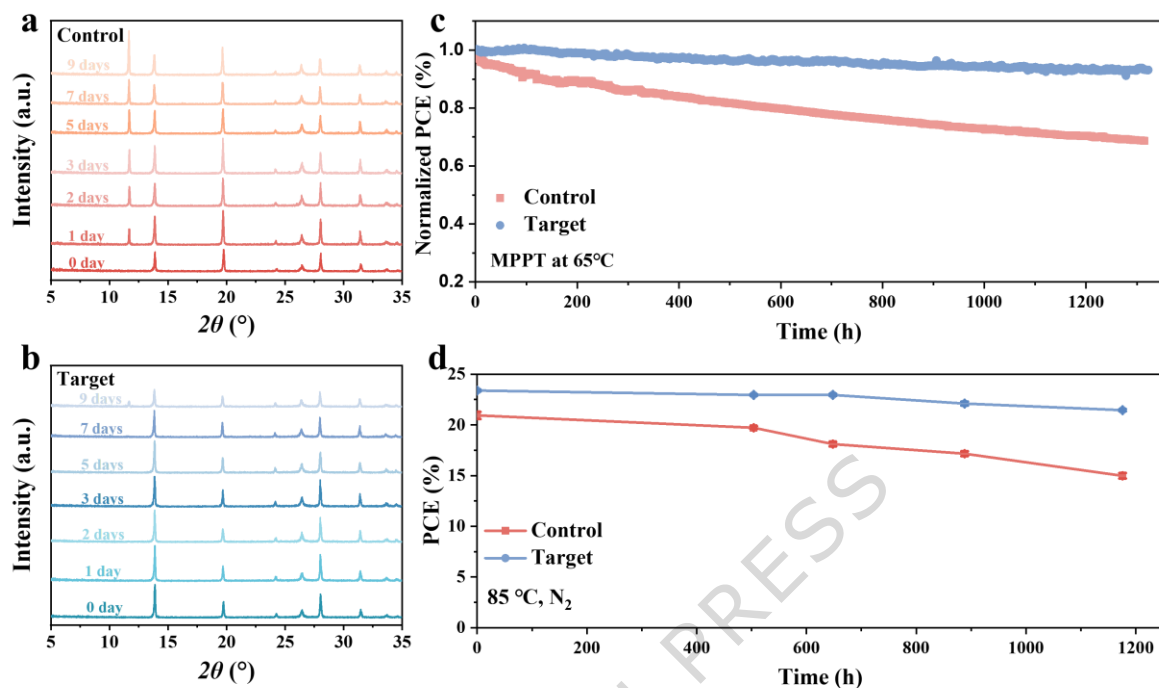


Fig. 5 | Stabilities of perovskite photovoltaics. XRD patterns of **a** control and **b** DPTA-treated perovskite films aging in air with > 60% RH. **c** Operational stabilities of the control and DPTA-treated perovskite devices tested at MPPT under continuous 1 sun illumination at 65 °C in glovebox. **d** Thermal stability of the unencapsulated perovskite devices with and without DPTA treatment at 85 °C. Error bars represent the standard deviation calculated from 10 independent devices.